Mini Project Report

Food Expiry Reminder App

1.Introduction

Food wastage and expired consumable products have become common problems in daily life. People often forget the expiry dates of food items, medicines, dairy products, and groceries stored at home, which can lead to health issues, financial loss, and unnecessary waste. Monitoring expiry dates manually becomes difficult when multiple products are stored for long periods.

This project focuses on developing a **Food Expiry Reminder App** using **Android Studio with Kotlin and Jetpack Compose**. The application helps users manage food and medicine expiry dates efficiently by storing item details and generating reminder notifications before expiration.

The system allows users to add food products, categorize them, track remaining days, and receive timely alerts about products nearing expiry. The application uses a modern Material 3 user interface with local database storage using Room Database for efficient data management.

Unlike traditional reminder systems, this application provides an intelligent and user-friendly mobile solution with expiry tracking, dashboard analysis, category management, search functionality, and notification alerts to reduce food wastage and improve health safety.

2.Objectives

The objectives of this project are:

- To develop a mobile application for tracking food and medicine expiry dates.
- To store product details such as item name, category, and expiry date using Room Database.
- To generate reminder notifications before product expiration.
- To reduce food wastage and improve household management.
- To provide an easy-to-use and modern mobile interface using Jetpack Compose.
- To categorize products such as milk, fruits, vegetables, medicines, and snacks.
- To display expired and expiring-soon products separately for better monitoring.
- To implement search and filtering functionality for quick product access.
- To create a dashboard showing total items, expired products, and expiring items.

3.Methodology

The methodology for this project involves the following steps:

- **Requirement Analysis** – The system requirements are analyzed to identify important features such as item management, expiry tracking, notifications, and dashboard statistics.
- **UI Design** – Modern user interfaces are designed using Jetpack Compose and Material 3 components to provide a responsive and user-friendly experience.
- **Database Creation** – Room Database is implemented to store food item details locally within the Android application.
- **Data Management** – Users can add, edit, delete, and search food items along with their expiry dates and categories.
- **Expiry Calculation** – The system calculates the remaining days between the current date and expiry date to determine item status.
- **Notification System** – Android AlarmManager and NotificationManager are integrated to send reminder notifications before product expiry.
- **Dashboard Implementation** – A dashboard screen is developed to display statistics such as total items, expired items, and products expiring soon.
- **Search and Filtering** – Search functionality and category-based filtering are implemented for efficient product management.
- **Dark Mode Support** – The application supports both light and dark themes using Material 3 design principles.
- **System Integration and Testing** – All modules including database, notifications, UI screens, and navigation are integrated and tested within Android Studio.

4.KOTLIN CODE FOR Food Expiry Reminder App

```
package com.example.foodexpiryreminder.database

import androidx.room.Entity
import androidx.room.PrimaryKey

@Entity(tableName = "expiry_items")
data class ExpiryItem(

    @PrimaryKey(autoGenerate = true)
    val id: Int = 0,
```

```
    val itemName: String,
    val category: String,
    val expiryDate: String,
    val notes: String = "",

    val isFavorite: Boolean = false,
    val isExpired: Boolean = false
)
```

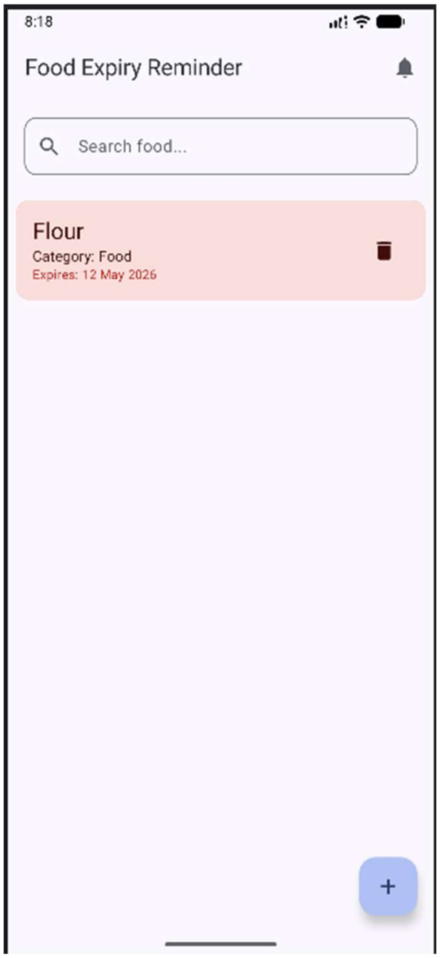
```
package com.example.foodexpiryreminder.notifications

import android.app.NotificationChannel
import android.app.NotificationManager
import android.content.Context
import android.os.Build
import androidx.core.app.NotificationCompat

object NotificationHelper {

    private const val CHANNEL_ID = "expiry_channel"

    fun createNotification(context: Context, itemName: String) {

        val manager =
            context.getSystemService(Context.NOTIFICATION_SERVICE)
                    as NotificationManager

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {

            val channel = NotificationChannel(
                CHANNEL_ID,
                "Expiry Notifications",
                NotificationManager.IMPORTANCE_HIGH
            )

            manager.createNotificationChannel(channel)
        }
```

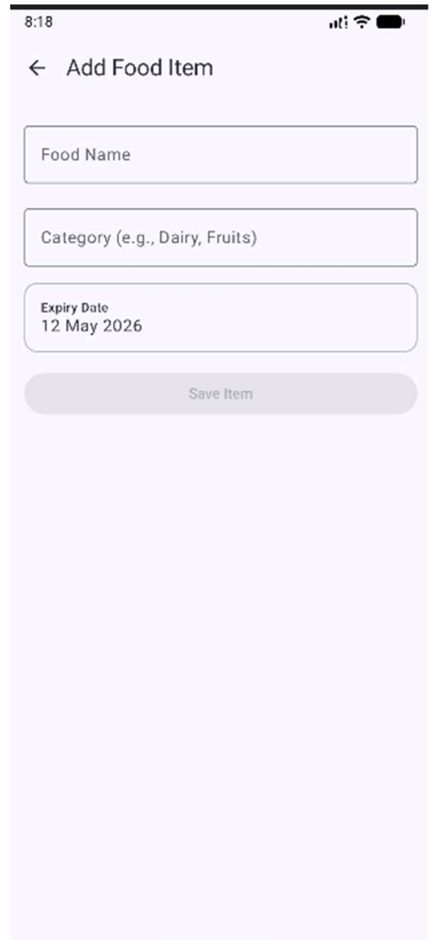
```
        val notification = NotificationCompat.Builder(context, CHANNEL_ID)
            .setContentTitle("Food Expiry Alert")
            .setContentText("$itemName is expiring soon!")
            .setSmallIcon(android.R.drawable.ic_dialog_alert)
            .build()

        manager.notify(itemName.hashCode(), notification)
    }
}
```

```
package com.example.foodexpiryreminder.screens

import androidx.compose.foundation.layout.*
import androidx.compose.foundation.lazy.LazyColumn
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp

@Composable
fun HomeScreen() {

    Scaffold(
        floatingActionButton = {
            FloatingActionButton(onClick = { }) {
                Text("+")
            }
        }
    ) { padding ->

        Column(
            modifier = Modifier
                .padding(padding)
                .padding(16.dp)
        ) {
```

```
            Text(
                text = "Food Expiry Reminder",
                style = MaterialTheme.typography.headlineMedium
            )

            Spacer(modifier = Modifier.height(16.dp))

            LazyColumn {

                items(5) {

                    Card(
                        modifier = Modifier
                            .fillMaxWidth()
                            .padding(vertical = 8.dp)
                    ) {

                        Column(
                            modifier = Modifier.padding(16.dp)
                        ) {

                            Text(text = "Milk")
                            Text(text = "Expires in 2 days")
                        }
                    }
                }
            }
        }
    }
}
```

5. Food Expiry Reminder App Output





6.Expected Results

- Accurate tracking of food and medicine expiry dates.
- Timely notification alerts before product expiration.
- Fast and responsive mobile application performance.
- Improved food management and reduced wastage.
- Easy product categorization and search functionality.
- Dashboard visualization of expired and active items.
- Integration of Room Database with Jetpack Compose UI.
- User-friendly and modern Android application experience.

7.Conclusion

This project demonstrates how Android application development can be used to solve real-life problems related to food wastage and product expiry management. The Food Expiry Reminder App successfully stores product information, tracks expiry dates, and generates timely notifications using Kotlin and Jetpack Compose.

By integrating Room Database, Material 3 UI, notification systems, and modern Android development practices, the application provides an intelligent and user-friendly solution for household food management. The dashboard, search functionality, and expiry tracking features further improve usability and effectiveness.

This project can be further enhanced by integrating barcode scanning, cloud synchronization, AI-based expiry prediction, Firebase authentication, and grocery shopping recommendations.

8.References

- Android Developers Documentation
- Kotlin Programming Documentation
- Jetpack Compose Documentation
- Room Database Documentation
- Material Design 3 Documentation
- NotificationManager Documentation
- AlarmManager Documentation
- Google Android Studio Documentation
- Android Architecture Components Guide
- Android Jetpack Libraries Documentation